Communication Protocol

for

Solterra K Series Low Voltage Charging Modules

Version 2.01

2024.09.27

Supported Power Modules

P/N	Description
ST120-10K	10kW low voltage charging module with output voltage range from 30Vdc to 120Vdc
STL120-06K	06kW low voltage charging module with output voltage range from 30Vdc to 120Vdc
Others	4.5kW, 10kW, 12kW, 14kW, 20kW and Laser Power Supplies

Revision History

Date	Version	Changes	By
2021-04-20	V1.0	First Releasing	Diane Chang
2022-09-19	V2.01	Adding Cautions	Tang Jianjun

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1 Information

This document describes the CAN-based communication protocol for communication between one or more charging modules and a charging controller.

Please carefully read the communication characteristics of the charging module described as follows to facilitate the use of it.

Here, the charging controller, as the superordinate controller, always handles communication with the bus devices on the linear CAN-BUS.

The CAN-based communication protocol supports the following communication methods:

- Peer-to-peer
- Multicast (if needed)
- Broadcast (if needed)

The information in this document is subject to change without prior notice.

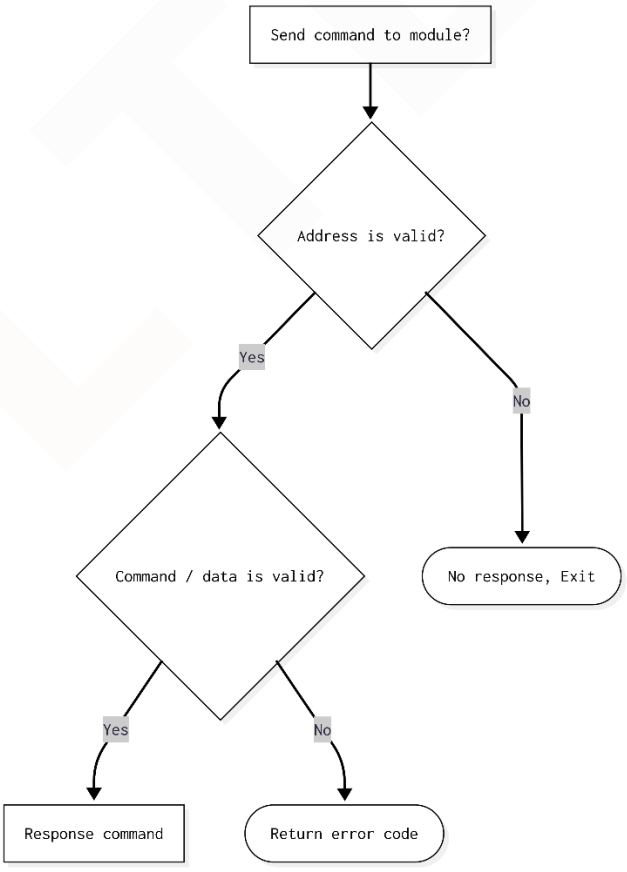
2 Summary

2.1 CAN Protocol

- Uses the extended frame mode CAN 2.0B
- The default bit rate is 125 Kbps (customizable)
- 29-bit identifier, extended frame

2.2 Communication Procedure

The following flowchart illustrates the standard communication procedure between the charging controller and a module:



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3 Frame Format

3.1 Frame Structure

The 29-bit identifier is defined as follows:

28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Byte3							Byte2								Byte1							Byte0						
Command ID_H															Module address													

3.2 Command Parse 1

The following table defines the command frame structure for Command Parse 1:

Name	Command ID	Module No.	Data Length	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
Monitor Setting Output	(0x1907,C080)>>7	0~60	8	CMD=0	Current (mA) × 1000			Voltage (mV) × 1000			
The module response output	(0x1807,C080)>>7	1~60	8	CMD=0	NZ,NZ: current, voltage set successfully 0,NZ: current failed, voltage succeeded NZ,0: current success, voltage failure 0,0: Current, voltage setting failure						
Request module information	(0x1907,C080)>>7	1~60	8	CMD=1	/	/	/	/	/	/	
Module Response Status	(0x1807,C080)>>7	1~60	8	CMD=1	NZ	Current (A×10)		Voltage (V×10)		Status1	Status0
Setting the module switching on and off	(0x1907,C080)>>7	0~60	8	CMD=2	/	/	/	/	/	/	0x55=Power on 0xAA=Power off
Module Response Switching	(0x1807,C080)>>7	1~60	8	CMD=2	NZ, NZ: setup successful 0, 0: Setting failure						

› **Address 0** indicates broadcast command, which is executed for all modules on the bus. The module does NOT reply to broadcast commands.

› **Addresses 1~60** indicate the module address; the corresponding module address replies to the data.

› **NZ** indicates non-zero data (default 0xFF). The data field transmits the high byte first, followed by the low byte.

› **Communication Timeout:** If the module does not receive a monitor command within 2 minutes, the communication is considered to be interrupted.

"/ " indicates current reservation, filled with 0.

Module Status Flag Bits: Byte7:

Byte7 : Bitmap	7	6	5	4	3	2	1	0
Define State 0	**1: Output undervoltage** 0: Normal output	**1: Output overvoltage** 0: Normal output	**1: Input undervoltage** 0: Input normal	**1: Input overvoltage** 0: Input normal	**1: Fan failure** 0: Fan normal	**1: Module constant current** 0: Module constant voltage	**1: Module failure** 0: Module normal	**1: Module shutdown** 0: Module running
Panel Indicator Lights	1: Yellow light on	1: Red light on	1: Red light on	1: Red light on	1: Red light flashing	1: Yellow light on	1: Red light on	

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Module Status Flag Bits: Byte6:

Byte6 : Bitmap	7	6	5	4	3	2	1	0
Define State 1	/	/	/	/	/	1: Setting up the shutdown 0: Setting on	1: Over-temperature protection 0: Module normal	1: Overcurrent protection 0: Module normal
Panel Indicator Lights	/	/	/	/	/	1: Green light flashing 0: Green light on	1: Red light on	1: Red light on

Note: The above alarms (in addition to the panel indicator) will cause the module to automatically protect and shut down when the indicator is red. In all other cases, the module outputs normally.

3.3 Command Parsing 2

The following table defines the command frame structure for Command Parsing 2:

Name	Command ID	Module No.	Data Length	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
Read monitoring set voltage	(0x1901,0080)>>7	1~60	0	empty							
The module response the voltage value	(0x1801,0080)>>7	1~60	8							Setting voltage (V × 10)	
Read Monitor Setting Current Limit	(0x1901,0880)>>7	1~60	0	empty							
The module response the current limit value	(0x1801,0880)>>7	1~60	8	Current limit value (A × 10)							
Read the input voltage value	(0x1907,A080)>>7	1~60	8	0x31	/	/	/	/	/	/	/
Module response input voltage	(0x1807,A080)>>7	1~60	8	0x31		Vab/32 (Single Phase Module Vac/32)		Vbc/32		Vca/32	
Read ambient temperature value	(0x1900,8080)>>7	1~60	0	empty							
Module response to ambient temperature	(0x1800,8080)>>7	1~60	8	/	/	/	/	Ambient temperature (°C × 10)		/	/
Module current sharing control	(0x19C2,1880)>>7	0	6	/	/	/	0x55 = Shutdown Module Auto Current Sharing 0xAA = Turn on module current sharing	/	/	/	/
Module internal current sharing	(0x1807,8080)>>7	1~60	8	The module automatically reports the current sharing data every 400ms; no need to pay attention to this mode.				/	/	/	/
Module Address Lookup	(0x19C2,2880)>>7	1~60	8	/	/	/	0x55 = Module lookup (Green light flashes 10s)	/	/	/	/

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4 Examples of Usage

4.1) Setting up the broadcasting on/off

Request ID	1907C080							
Data	02	00	00	00	00	00	00	55
Response ID	Not applicable							
Data	empty							
Response	No return from broadcast command							

4.2) Setting up module1# switching on/off

Request ID	1907C081							
Data	02	00	00	00	00	00	00	55
Response ID	1807C081							
Response Data	02	FF	00	00	00	00	00	00
Note	Byte1 returns FF = success, 00 = failure							

4.3) Setting the output voltage and current of module3#

Request ID	1907C083							
Data	00	01	5F	90	00	01	86	A0
Response ID	1807C083							
Response Data	00	FF	00	00	00	00	00	00
Calculation	Output = 100V, 90A 100V = 100 × 1000 = 100,000 mV = 0x186A0 90A = 90 × 1000 = 90,000 mA = 0x15F90							
Note	Byte1: returns FF = success, 00 = failure							

4.4) Read the status information of module2#

Request ID	1907C082							
Data	01	00	00	00	00	00	00	00
Response ID	1807C082							
Response Data	01	00	01	17	03	E8	00	00
Output Voltage	Byte4~Byte5: 0x03E8 = 1000 / 10 = 100V							
Output Current	Byte2~Byte3: 0x0117 = 279 / 10 = 27.9A							
Output Status	Byte6~Byte7: 0x0000 = Normal operation, no alarms							

4.5) Read the output voltage value of module1# for monitoring setting

Request ID	19010081							
Data	(empty)							
Response ID	18010081							
Response Data	01	00	18	36	00	1E	03	84
Output Voltage	Byte6~Byte7: 0x0384 = 900 / 10 = 90V							

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4.6) Read and monitor the output current limit value of module5#

Request ID	19010885							
Data	(empty)							
Response ID	18010885							
Response Data	01	3D	00	00	00	00	40	2C
Current Limit	Byte0~Byte1: 0x013D = 317 / 10 = 31.7A							

4.7) Read the input voltage value of module1#

Request ID	1907A081							
Data	31	00	00	00	00	00	00	00
Response ID	1807A081							
Response Data	31	01	2F	1E	2F	2C	2F	26
Vab	Byte2~Byte3: 0x2F1E = 12062 / 32 = 376.9V							
Vbc	Byte4~Byte5: 0x2F2C = 12076 / 32 = 377.4V							
Vca	Byte6~Byte7: 0x2F26 = 12070 / 32 = 377.2V							

4.8) Read the ambient temperature of module1#

Request ID	19008081							
Data	(no data)							
Response ID	19008081							
Response Data	00	1F	0B	B5	01	52	03	A1
Ambient Temp	Byte4~Byte5: 0x0152 = 338 / 10 = 33.8°C							

4.9) Close the module autonomous equalization of flow

Request ID	19C21880							
Data	00	00	00	55	00	00		
Response	No response							
Note	Byte4 = 0x55: Turn equalization OFF Byte4 = 0xAA: Turn equalization ON							

4.10) Module address search

Issued ID	19C22882							
Data	00	00	00	55	00	00	00	00
Return ID	19C22882							
Return Data	(no data)							
Result	Module at address 2 blinks green LED for 10 seconds							

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5 Cautions

5.1) Soft Start Loading

The module cannot be directly loaded during the soft start process. It must be loaded after the output voltage of the module has been established (output steady state), which can be judged by the module status flag bit.

Table 5-1: Module Status Flag Bits (Byte7)

Byte7 : Bit	7	6	5	4	3	2	1	0
Define State 0	**1: Output undervoltage** 0: Normal output	**1: Output overvoltage** 0: Normal output	**1: Input undervoltage** 0: Input normal	**1: Input overvoltage** 0: Input normal	**1: Fan failure** 0: Fan normal	**1: Module constant current** 0: Module constant voltage	**1: Module failure** 0: Module normal	**1: Module shutdown** 0: Module running
Panel Indicator Lights	1: Yellow light on	1: Red light on	1: Red light on	1: Red light on	1: Red light flashing	1: Yellow light on	1: Red light on	

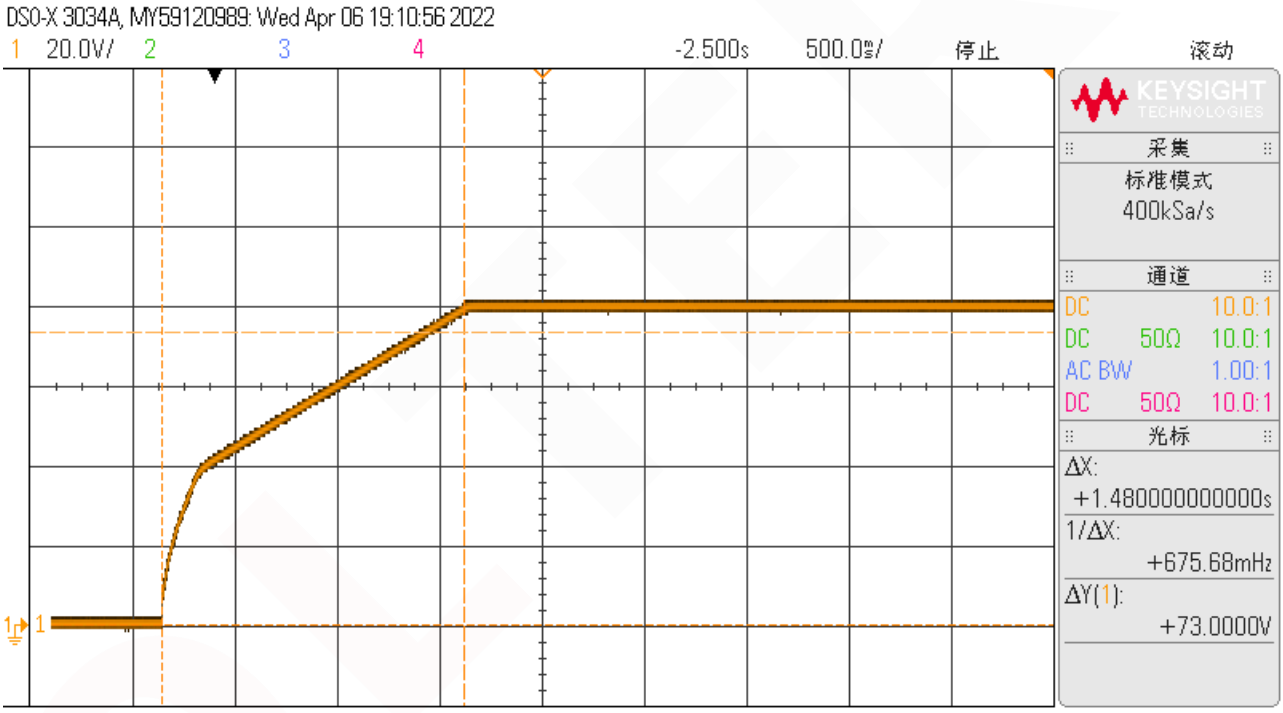


Figure 5-1 Description: The oscilloscope trace shows the soft start ramp phase followed by steady output. The module reaches steady output after approximately 1.5 seconds of soft start. The output voltage rises gradually from 0V during the soft start phase, reaching approximately +73V at steady state. Monitor the status flag (Byte7 Bit0 = 0: Module running) before applying load.

5.2) Current Ramp-Up Procedure

The output current of the module should be increased in a stepwise manner. For example, if the demand current is 100A:

- Set the initial current value to 10A (10% of the target load current).
- Increase in steps of 10%: 10A, 20A, 30A, ..., 90A, 100A.
- The recommended current step interval (creepage time) is ≥ 1 second per step.

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6 Baud Rate Configuration

With CAN communication functioning properly, set the module address to 1. Send the following command to configure the baud rate. The new setting will take effect after reset or restarting the power module.

Extended Frame ID	DATA(0)	DATA(1)	DATA(2)	DATA(3)	DATA(4)	DATA(5)	Baud Rate
0x1907A081	29	0	0	62	FF	FF	125K
0x1907A081	29	0	0	62	55	55	250K
0x1907A081	29	0	0	62	AA	AA	500K

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